a Genealogy cohorts: worlds and nested lines

Cohort	Worlds	Confined worlds	Lines	Open-first lines
N3-v6	20	17	1088	1088
N6	20	20	1280	1280

N3-v6 = discovery; N6 = prospective confirmation

Open-first: earliest copier on each sampled ancestry has partner fetch

World = simulation; line = nested sampled-cell ancestry
Confined worlds have a first confined founder

b First founders versus later conversions

Cohort	First founders	Later conversions	Events
N3-v6	17	15	32
N6	20	14	34

20 worlds/cohort; at most one first founder/world

Later conversion events can repeat within a world

First founders and subsequent conversion events stay separate

c N6 first-founder mechanisms

Recorded mechanism	Founders
novel	18
damage	2

novel = rewrite, <75% match to either parent (18)

damage = partial overwrite, ≥75% parent match (2)

Separate N6-1 source rule: copy/rewrite 18/20, mutation 0/20
Recorded event categories and causal source classes differ

d Functional donors: two definitions

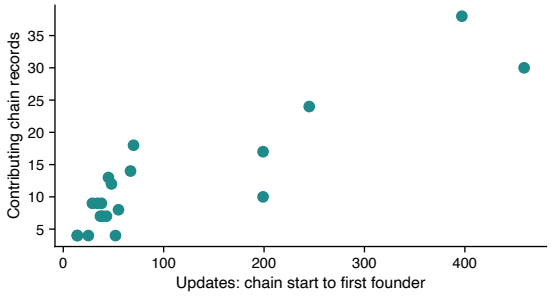
Criterion	Whole-tape	Executed bytes
Non-copier	20	11
Copier	0	9

Executed-byte non-copiers: 11/20

All whole-tape noncopier executors vs soup-abundance null: P=0.062

20 N6 executors; whole tape vs only instruction-stream bytes
Copier: ≥75% scoped bytes copied in ≥50% partner encounters
N6-4 needs 12/20 executed-byte noncopiers; 11/20 observed: not met

e N6: all first-founder chain lengths



N6: one point per first founder, 20 worlds
Records = contributing chain events; intervening whole-tape donors are non-copiers (executed-byte function kept separate)

f Registered N6: complete component ledger

ID	Component	Observed	Verdict
N6-1	Copy/rewrite; mutation	18/20; 0/20	met
N6-2	Whole-tape noncopier chains	20/20	met
N6-3	Executed-word share	Observed 0.00190 7.9× baseline	met
N6-4	Whole/exec-byte noncopiers	20/20; 11/20 P = 0.062	not met
N6-5′	Expected mutation founders	0.0064 in cohort	met

N6-1 copy/rewrite ≥80%, mutation ≤10%; N6-2 whole-chain ≥75%
N6-3 median cells carrying executed word ≥5×random baseline 0.00024
N6-5′ expected mutation founders <0.05/cohort; ≥25% mutation rejects